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Load the array J

```
load modball
```

Display J, properly scaled

notes im222 expects input scaled to [0,1], so J must be divided by 255.

```
figure;
im222(double(J)/255);
title('J as given (wrapped/''solarized'' by the mod-256 shift)');
```



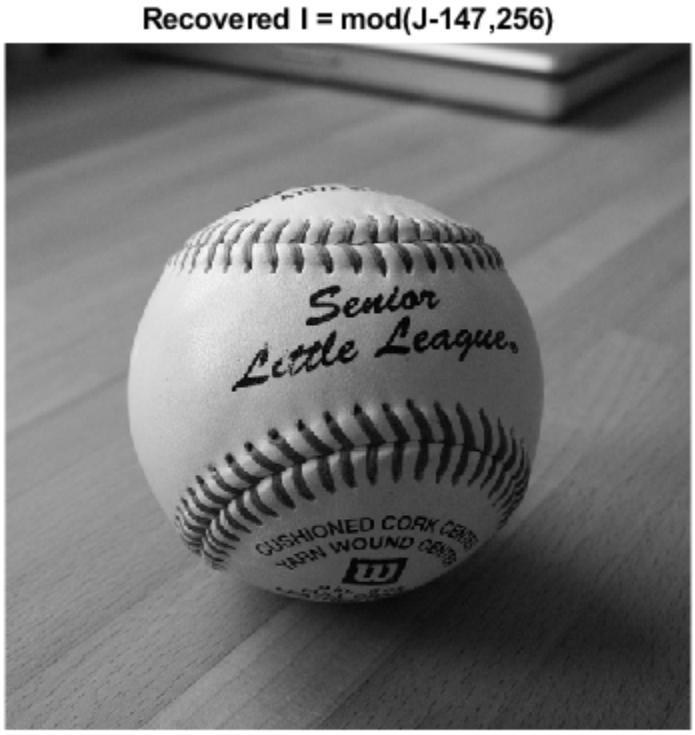
Recover the original image I from J

explains why mod(J-147,256) works: mod-256 addition is a bijection on 0–255 so it's fully invertible;flags that line as the key step.

```
I = mod(J - 147, 256);
```

Display and visually verify

```
figure;
im222(double(I)/255);
title('Recovered I = mod(J-147,256)');
% Visual check confirms a sharp photograph is recovered
% (no tearning like in the J display above), which
% confirms the inversion formula is correct.
```



Save the reconstructed image

```
imwrite(uint8(I), 'recovered_image.png');
```

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